# NISM V-D Derivatives Summary

All 10 derivatives chapters, 79 subtopics, in tables, formulas and payoff charts. Every subtopic is rated for exam importance. Derivatives = 55% of the exam. Read 5★ first.

★★★★★ must know, asked often ★★★★☆ high ★★★☆☆ medium ★★☆☆☆ low ★☆☆☆☆ skim

‡ rule or number that changes often; check workbook edition

| Ch | Chapter | Marks | Priority | Subtopics | 5★ |
|---|---|---|---|---|---|
| 13 | Basics of Derivatives | 10 | ★★★★★ | 12 | 4 |
| 14 | Understanding Index | ≈4–6 | ★★★☆☆ | 7 | 1 |
| 15 | Forwards & Futures | 15 | ★★★★★ | 12 | 7 |
| 16 | Introduction to Options | 13 | ★★★★★ | 11 | 6 |
| 17 | Strategies using Futures & Options | ≈8–10 | ★★★★★ | 7 | 4 |
| 18 | Interest Rates, Instruments & Fixed Income Markets | ≈6 | ★★★★☆ | 10 | 3 |
| 19 | Interest Rate Derivatives (OTC) | 2 | ★★☆☆☆ | 4 | 0 |
| 20 | Exchange Traded Interest Rate Futures | 10 | ★★★★★ | 8 | 2 |
| 21 | Exchange Traded Interest Rate Options | ≈5 | ★★★☆☆ | 4 | 0 |
| 22 | Strategies using Interest Rate Derivatives | ≈5 | ★★★★☆ | 4 | 2 |

### 5★ list (29) — read these first

**Ch 13** · Types of derivatives · Exchange-traded vs OTC · Participants · Margins & risk management
**Ch 14** · Weighting methods
**Ch 15** · Futures: features · Futures terminology · Futures payoff · Pricing: cost of carry · Basis, contango, backwardation, convergence · Uses: hedge, speculate, arbitrage · Beta hedging
**Ch 16** · Option basics · Moneyness · Intrinsic & time value · Payoff profiles · Factors affecting premium · Option Greeks
**Ch 17** · Hedging strategies · Spread strategies · Volatility strategies · Strategy selection matrix
**Ch 18** · Bond pricing · Yield measures · Duration, PVBP, convexity
**Ch 20** · CTD, conversion factor, invoice price · Hedging with IRF
**Ch 22** · Hedging & duration management · Yield curve strategies

### 4★ list (25)

**Ch 13** · Definition & underlying · History — India · Clearing & settlement
**Ch 14** · Index construction · Free float · Desirable attributes & impact cost
**Ch 15** · Forward contracts · Open interest & volume · Contract specifications (India)
**Ch 16** · Put-call parity · Volatility · Options vs futures · Contract specs & margins (India)
**Ch 17** · Synthetic positions · Directional trades
**Ch 18** · Interest rate basics · Monetary policy tools (RBI) · Money market instruments · Bond features · Yield curve & term structure
**Ch 20** · Contract features · Pricing of bond futures · Speculation & arbitrage
**Ch 21** · Features · Payoffs & uses

Built from the NISM Series VIII (equity derivatives) and interest rate derivatives syllabi that V-D draws on. Marks: Ch13 10, Ch15 15, Ch16 13, Ch19 2, Ch20 10 from published tables; others estimated. Not copied from the workbook.

# 13 Basics of Derivatives

≈10 marks · 12 subtopics · chapter priority ★★★★★

## Definition & underlying ★★★★☆

| | |
|---|---|
| **Derivative** | Contract whose value is derived from an underlying |
| **Underlying** | Shares, index, commodity, currency, interest rate, bond, credit, weather… |
| **Legal (India)** | SCRA 1956 Sec 2(ac): derivative = security (amended 1999) |
| **Key features** | Leverage · zero-sum (before costs) · future settlement · notional value |

Notional value = price × lot × lots

Zero-sum: one side's gain = other side's loss

## Types of derivatives ★★★★★

| Type | What | Traded |
|---|---|---|
| Forward | Agreement to buy/sell later at price fixed today | OTC |
| Futures | Standardised forward, daily MTM | Exchange |
| Option | Right (not obligation) to buy (call) / sell (put) | Both |
| Swap | Exchange of cash-flow streams (e.g. fixed ↔ floating) | OTC |
| Others | Warrants, LEAPS (long-dated options), basket options, swaptions | Varies |

## Exchange-traded vs OTC ★★★★★

| Feature | Exchange-traded | OTC |
|---|---|---|
| Terms | Standardised | Customised |
| Counterparty risk | None to traders (clearing corp, novation) | Yes |
| Margins / MTM | Mandatory, daily | Usually none / bilateral |
| Liquidity | High | Low |
| Transparency | Public prices | Private |
| Regulation | SEBI (exchange) | RBI for OTC rate/FX |
| Examples | Index futures, stock options, IRF | Forwards, IRS, FRA |

## Participants ★★★★★

| Participant | Goal | Example |
|---|---|---|
| Hedger | Reduce existing risk | Fund sells index futures to protect portfolio |
| Speculator | Profit from view; takes risk; gives liquidity | Buys futures expecting a rise |
| Arbitrageur | Riskless profit from mispricing; aligns prices | Buy spot, sell overpriced futures |
| Margin trader | Uses leverage of margin | Buys futures instead of cash shares |

## Economic functions ★★★☆☆

Price discovery · Risk transfer (hedgers → speculators)

More liquidity in cash market · Lower transaction cost

Market completeness · Encourages entrepreneurial activity

**TRAP** Derivatives do NOT guarantee returns or create wealth.

## History — world ★★☆☆☆

Rice futures, Dojima Japan (1700s)

Chicago Board of Trade (CBOT) 1848 — first organised futures exchange

Currency futures CME 1972 · CBOE options 1973

Black-Scholes model 1973

## History — India ★★★★☆

| Year | Event |
|---|---|
| 1996–98 | L.C. Gupta Committee — framework for derivatives trading |
| 1998 | J.R. Varma Committee — risk containment (margins) |
| 1999 | SCRA amended — derivatives = securities |
| Jun 2000 | Index futures (first) |
| Jun 2001 | Index options |
| Jul 2001 | Stock options |
| Nov 2001 | Stock futures |
| 2008 | Currency futures |
| 2009/2014 | IRF relaunch / cash-settled bond futures ‡ |
| 2018–19 | Physical settlement of stock F&O |

Badla (carry-forward) system banned 2001

## Clearing & settlement ★★★★☆

| | |
|---|---|
| **Clearing corporation** | Central counterparty: buyer to every seller, seller to every buyer (novation) |
| **Members** | Trading member (TM) · Clearing member (CM) · Self-clearing member · Professional clearing member (clears for others, doesn't trade) |
| **Settlement Guarantee Fund (SGF)** | Covers member default so settlement completes |
| **MTM settlement** | Daily, in cash (T+1) |
| **Final settlement** | Index: cash · Stocks: physical delivery |

## Margins & risk management ★★★★★

| Margin | Rule |
|---|---|
| Initial margin | SPAN (scenario worst-case loss, portfolio-based, VaR) — paid upfront |
| Exposure / ELM margin | Extra flat % over SPAN |
| MTM | Daily P&L in cash |
| Premium | Option buyer pays full premium upfront |
| Delivery margin | Near expiry for physically settled positions |
| Who pays | Futures: both sides · Options: seller only |

Position limits: client, trading member, market-wide

MWPL: stock enters ban at 95% OI; exits below 80%

Peak margin reporting ‡

## Risks in derivatives ★★★☆☆

Leverage risk · Market risk · Counterparty risk (OTC) · Liquidity risk

Basis risk · Operational / legal risk · Model risk

### Regulation & stock eligibility ★★☆☆☆

- SCRA 1956, SEBI Act 1992, SEBI regulations, exchange bye-laws
- Stocks chosen by market cap, liquidity, MWPL, median order size ‡
- Min contract value for index derivatives ₹15–20 lakh (Nov 2024) ‡
- One weekly index expiry per exchange ‡

### Taxation of derivatives ★★★☆☆

| Item | Rule ‡ |
|---|---|
| F&O income | Non-speculative business income (Sec 43(5)) |
| Losses | Set off vs business income; carry forward 8 years |
| STT futures | On sell side, % of value (raised Oct 2024; check latest) |
| STT options | On premium (sell); on intrinsic value if exercised |
| Audit | Turnover-based tax audit rules |

## 14 Understanding Index

≈4–6 marks · 7 subtopics · chapter priority ★★★☆☆

### What an index is & uses ★★★☆☆

- Barometer of market
- Benchmark for funds
- Underlying for index funds, ETFs, futures, options
- Measures relative change from base

### Index construction ★★★★☆

```
Index value = Current market cap ÷ Base market cap × Base value
(Free-float market cap = Price × Shares × Investible weight factor)
Divisor adjusted for corporate actions so the index doesn't jump
```

| | |
|---|---|
| **Base date / value** | Starting point (e.g. Nifty 1,000 on 3 Nov 1995) |
| **Selection** | Liquidity (impact cost), free float, market cap, listing history |
| **Rebalancing** | Semi-annual review; replacement on delisting/merger |

### Weighting methods ★★★★★

| Method | Weight by | Example / note |
|---|---|---|
| Free-float market cap | Price × public float shares | Nifty 50, Sensex (standard in India) |
| Full market cap | Price × all shares | Older method |
| Price-weighted | Share price only | Dow Jones; high-price stock dominates |
| Equal-weighted | Same weight | Nifty 50 Equal Weight; needs frequent rebalancing |
| Factor / fundamental | Earnings, value, momentum etc. | Smart-beta indices |

> **TRAP** In a market-cap index, a split or bonus does NOT change the index (market cap unchanged). Rights/new issues do → divisor changes.

### Free float ★★★★☆

| | |
|---|---|
| **Free float** | Shares available for public trading |
| **Excludes** | Promoter holdings, govt strategic holding, locked-in shares, cross-holdings, FDI strategic stakes |
| **Investible weight factor (IWF)** | Free-float shares ÷ total shares |

### Major Indian indices ★★★☆☆

| Index | Stocks | Base | Base value |
|---|---|---|---|
| Nifty 50 (NSE) | 50 | 3 Nov 1995 | 1,000 |
| BSE Sensex | 30 | 1978–79 | 100 |
| Nifty Bank | Banking stocks | 1 Jan 2000 | 1,000 |
| Nifty Next 50 | 50 | | |
| Nifty Midcap 150 / Smallcap 250 | 150 / 250 | | |

### Desirable attributes & impact cost ★★★★☆

```
Ideal price  = (best bid + best ask) ÷ 2
Impact cost  = (actual execution price - ideal price) ÷ ideal price × 100
Lower impact cost = more liquid
```

- Diversified (low unsystematic risk)
- Liquid (low impact cost)
- Professionally maintained
- Transparent rules

### Index-based products & uses ★★★☆☆

- Index funds
- ETFs (traded on exchange)
- Index futures & options
- Benchmarking
- Beta hedging
- Tracking error = SD of return difference

## 15 Forwards & Futures

≈15 marks · 12 subtopics · chapter priority ★★★★★

### Forward contracts ★★★★☆

| | |
|---|---|
| **Features** | OTC, bilateral, customised size/date, settled at maturity |
| **Limitations** | Counterparty risk · illiquid · no price transparency · hard to exit |
| **Use** | Exact-date hedges (e.g. exporter with bank) |

### Futures: features ★★★★★

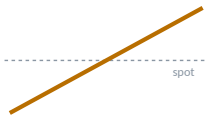
- Standardised (lot, expiry, quality)
- Exchange-traded
- Clearing corp guarantee
- Daily MTM
- Both sides post margin
- Can close before expiry by opposite trade
- Linear payoff

## Futures terminology ★★★★★

| Term | Meaning |
|---|---|
| Spot price | Current cash-market price |
| Futures price | Agreed price for future date |
| Contract cycle | 3 months: near, mid, far |
| Expiry day | Last trading day of contract ‡ |
| Lot / contract size | Fixed quantity per contract |
| Basis | Spot − Futures |
| Cost of carry | Interest − income (dividend) |
| Initial margin | Upfront deposit (SPAN + exposure) |
| MTM | Daily settlement of P&L |
| Open interest | Outstanding contracts |
| Settlement price | Price used for MTM |

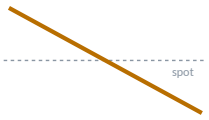
## Futures payoff ★★★★★

**Long futures**




Gain if price ↑, unlimited both ways

**Short futures**




Gain if price ↓

**Long spot + short futures**




Hedged: flat

Linear and symmetric · P&L = (exit − entry) × lot × lots

## Pricing: cost of carry ★★★★★

```
Simple:       F = S × (1 + r × t)
Continuous:   F = S × e^(r t)
With dividend: F = (S - PV(D)) × e^(r t)   or  F = S × e^((r - d) t)
Commodities:  F = S × e^((r + storage - convenience yield) t)

Ex: S 1,000, r 8%, 3 months → simple 1,020 · continuous 1,020.20
```

Fair value ignores views about future price · Higher r → higher F · Higher dividend → lower F

## Basis, contango, backwardation, convergence ★★★★★

| Term | Condition |
|---|---|
| Contango (normal) | F > S, basis negative |
| Backwardation | F < S, basis positive (dividend > interest) |
| Convergence | F → S as expiry nears; basis = 0 at expiry |
| Basis strengthens | S − F rises → good for long spot + short futures |

## Uses: hedge, speculate, arbitrage ★★★★★

| Situation | Trade |
|---|---|
| Hold shares, fear fall | Sell futures (short hedge) |
| Will buy later, fear rise | Buy futures (long / anticipatory hedge) |
| Bullish with limited cash | Buy futures (leverage) |
| Futures overpriced | Cash-and-carry: buy spot + sell futures |
| Futures underpriced | Reverse cash-and-carry: sell spot + buy futures |

Arbitrage funds earn ≈ money-market rate

## Beta hedging ★★★★★

```
Lots to hedge fully  = Portfolio value × β ÷ (Futures price × lot)
Change beta β₀ → β₁  = (β₁ - β₀) × Portfolio ÷ (Futures price × lot)   (-) sell · (+) buy

Ex: ₹1.5 cr, β 1.25, Nifty 25,000, lot 75 → 1.875 cr ÷ 18.75 L = 10 lots (sell)
```

Fully hedged portfolio earns ≈ risk-free rate · Hedge removes market risk, not stock-specific risk

## Open interest & volume ★★★★☆

| Price | OI | Reading |
|---|---|---|
| ↑ | ↑ | Long build-up (bullish) |
| ↓ | ↑ | Short build-up (bearish) |
| ↑ | ↓ | Short covering |
| ↓ | ↓ | Long unwinding |

Volume = contracts traded today; OI = contracts still open

## Spreads & rollover ★★★☆☆

Calendar spread: buy one expiry, sell another (same underlying) — lower margin

Inter-commodity spread: two related underlyings

Rollover: close near month, open next month; cost = spread

## Contract specifications (India) ★★★★☆

| Item | Index futures | Stock futures |
|---|---|---|
| Settlement | Cash | Physical delivery |
| Cycle | 3 monthly | 3 monthly |
| Final settlement price | Closing index value on expiry | Closing price on expiry |
| Lot | Min contract value ₹15–20 lakh ‡ | Set per stock ‡ |
| MTM | Daily | Daily |

## Basis risk & cross hedge ★★★☆☆

Basis risk: hedge closed before expiry or asset ≠ futures underlying

Cross hedge: e.g. mid-cap portfolio with Nifty futures

Perfect hedge only with same asset held to expiry

# 16 Introduction to Options

≈13 marks · 11 subtopics · chapter priority ★★★★★

## Option basics ★★★★★

| Term | Meaning |
|---|---|
| Call | Right to BUY at strike |
| Put | Right to SELL at strike |
| Buyer (holder) | Pays premium; has right |
| Seller (writer) | Gets premium; has obligation; pays margin |
| Strike (K) | Exercise price |
| Premium | Price of option |
| European | Exercise only on expiry (India: index & stock options) |
| American | Exercise any time till expiry |

## Moneyness ★★★★★

| | Call | Put |
|---|---|---|
| ITM | S > K | S < K |
| ATM | S = K | S = K |
| OTM | S < K | S > K |

**TRAP** Put with K 1,100 when S 1,050 is ITM by 50.

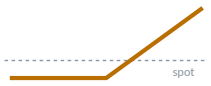
## Intrinsic & time value ★★★★★

```
Call IV = max(0, S - K)        Put IV = max(0, K - S)
Premium = Intrinsic value + Time value
OTM/ATM: premium = 100% time value · at expiry time value = 0
Ex: S 520, call K 500 @ 32 → IV 20, TV 12
```

Time value highest ATM | Time value decays faster near expiry

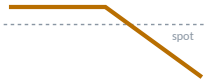
## Payoff profiles ★★★★★

**Long call**



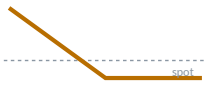

Max loss premium · BE K+P

**Short call**



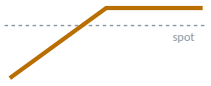

Max gain premium · loss unlimited

**Long put**



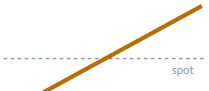

Max gain K−P · BE K−P

**Short put**



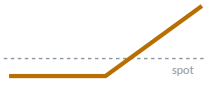

Max gain premium · max loss K−P

**Long futures (compare)**




Linear

**Protective put**




Stock + long put = call shape

| Position | Max profit | Max loss | Breakeven |
|---|---|---|---|
| Long call | Unlimited | Premium | K + P |
| Short call | Premium | Unlimited | K + P |
| Long put | K − P | Premium | K − P |
| Short put | Premium | K − P | K − P |

## Factors affecting premium ★★★★★

| Factor ↑ | Call | Put |
|---|---|---|
| Spot price | ↑ | ↓ |
| Strike price | ↓ | ↑ |
| Volatility | ↑ | ↑ |
| Time to expiry | ↑ | ↑ (usually) |
| Interest rate | ↑ | ↓ |
| Dividend | ↓ | ↑ |

## Option Greeks ★★★★★

| Greek | Measures | Long call | Long put | Key point |
|---|---|---|---|---|
| Delta | Δ premium per ₹1 spot | 0 to +1 | −1 to 0 | ATM ≈ ±0.5; = hedge ratio |
| Gamma | Δ delta per ₹1 spot | + | + | Highest ATM near expiry; same for call & put |
| Theta | Δ premium per day | − | − | Time decay; sellers gain |
| Vega | Δ premium per 1% vol | + | + | Highest ATM, long-dated |
| Rho | Δ premium per 1% rate | + | − | Least important |

```
Δ premium ≈ delta × Δspot   ·   new delta ≈ delta + gamma × Δspot
Ex: delta 0.6, spot +10 → premium +6
```

## Pricing models ★★★☆☆

| Model | Points |
|---|---|
| Binomial | Price moves up/down each step; works for American |
| Black-Scholes (1973) | European; inputs S, K, t, r, σ (+ dividend); assumes lognormal prices, constant σ and r, no costs |

Only σ is not directly observable

Implied volatility = σ backed out from market premium

## Put-call parity ★★★★☆

```
C + K·e^(-rt) = P + S        (European, same K and expiry)
Ex: C 50, S 1,000, PV(K) 980 → P = 50 + 980 - 1,000 = 30
Synthetic long futures = long call + short put
```

## Volatility ★★★★☆

| | |
|---|---|
| **Historical volatility** | From past prices (SD of returns, annualised) |
| **Implied volatility (IV)** | From current option prices; market's expectation |
| **India VIX** | NSE volatility index from Nifty option IVs; 'fear gauge' |
| **Put-call ratio (PCR)** | Put OI ÷ call OI — sentiment indicator |

### Options vs futures ★★★★☆

| | Futures | Options (buyer) |
|---|---|---|
| Obligation | Both sides | Buyer: right only |
| Upfront | Margin | Premium |
| Payoff | Linear | Non-linear |
| Max loss | Unlimited | Premium |

### Contract specs & margins (India) ★★★★☆

| Item | Index options | Stock options |
|---|---|---|
| Style | European | European |
| Settlement | Cash | Physical delivery |
| Premium settlement | T+1 | T+1 |
| Margin | Seller: SPAN + exposure · buyer: premium | Same |
| Exercise | Auto-exercise of ITM at expiry | Same |

STT on exercised options: on intrinsic value ‡

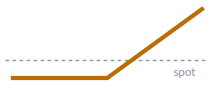
# 17 Strategies using Futures & Options

≈8–10 marks · 7 subtopics · chapter priority ★★★★★

### Hedging strategies ★★★★★

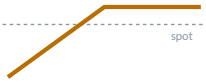
| Strategy | Build | Effect |
|---|---|---|
| Protective put | Long stock + long put | Floor on loss; keeps upside; costs premium |
| Covered call | Long stock + short call | Extra income; caps upside |
| Collar | Long stock + long put + short call | Low-cost floor and cap |
| Short futures hedge | Long stock + short futures | Locks value (both ways) |

**Protective put**




= long call shape

**Covered call**




= short put shape

**Collar**




Floor + cap

### Spread strategies ★★★★★

| Spread | Build | View | Net | Max profit | Max loss |
|---|---|---|---|---|---|
| Bull call | Buy low-K call, sell high-K call | Mod. bullish | Debit | $K_2$–$K_1$–net | Net debit |
| Bull put | Sell high-K put, buy low-K put | Mod. bullish | Credit | Net credit | $K_2$–$K_1$–credit |
| Bear put | Buy high-K put, sell low-K put | Mod. bearish | Debit | $K_2$–$K_1$–net | Net debit |
| Bear call | Sell low-K call, buy high-K call | Mod. bearish | Credit | Net credit | $K_2$–$K_1$–credit |

**Bull call spread**




BE = $K_1$ + net debit

**Bear put spread**




BE = $K_2$ − net debit

**Bull put spread**

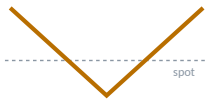



Credit

### Volatility strategies ★★★★★

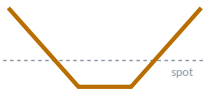
| Strategy | Build | View | Breakevens |
|---|---|---|---|
| Long straddle | Buy call + put, same K | Big move either way | K ± total premium |
| Short straddle | Sell call + put, same K | No move; unlimited risk | K ± total premium |
| Long strangle | Buy OTM call + OTM put | Big move; cheaper | K_put − prem, K_call + prem |
| Short strangle | Sell OTM call + OTM put | Range-bound | Same |
| Long butterfly | Buy 1 $K_1$, sell 2 $K_2$, buy 1 $K_3$ (calls) | Price stays near $K_2$; limited risk | $K_1$+debit, $K_3$−debit |

**Long straddle**




Same strike

**Long strangle**




Two strikes

**Long butterfly**




Low volatility

**TRAP** Straddle = same strike · strangle = different strikes.

### Synthetic positions ★★★★☆

| Synthetic | Equals |
|---|---|
| Long call + short put (same K) | Long futures |
| Short call + long put | Short futures |
| Long stock + long put | Long call |
| Long stock + short call | Short put |
| Long futures + long put | Long call |

### Strategy selection matrix ★★★★★

| View ↓ / Volatility → | Expect ↑ volatility | Expect ↓ volatility |
|---|---|---|
| Bullish | Long call, bull call spread | Short put, bull put spread |
| Bearish | Long put, bear put spread | Short call, bear call spread |
| Neutral | Long straddle / strangle | Short straddle / strangle, butterfly |

### Arbitrage strategies ★★★☆☆

- Cash-and-carry (spot vs futures)
- Conversion: long stock + long put + short call (parity)
- Reversal: short stock + short put + long call
- Box spread: bull call + bear put = riskless ($K_2$–$K_1$)

### Directional trades ★★★★☆

- Bullish: long futures / long call / short put
- Bearish: short futures / long put / short call
- Futures: linear, margin both sides
- Long options: limited loss, time decay

# 18 Interest Rates, Instruments & Fixed Income Markets

≈≈6 marks · 10 subtopics · chapter priority ★★★★☆

### Interest rate basics ★★★★☆

```
Simple:     FV = P × (1 + r t)
Compound:   FV = P × (1 + r/m)^(m t)
Continuous: FV = P × e^(r t)
Effective annual rate = (1 + r/m)^m - 1
PV = FV ÷ (1 + r)^t
Real rate ≈ nominal - inflation
```

- Time value of money
- Discounting
- Rule of 72

### Monetary policy tools (RBI) ★★★★☆

| Tool | Meaning |
|---|---|
| Repo rate | RBI lends to banks against G-secs (policy rate) |
| SDF | Standing Deposit Facility — floor of corridor |
| MSF | Marginal Standing Facility — ceiling, penal borrowing |
| Reverse repo | Banks park with RBI (now below SDF) |
| CRR | Cash reserve with RBI (% of deposits) |
| SLR | Liquid assets (G-secs) held by banks |
| OMO | RBI buys/sells G-secs to manage liquidity |

- Corridor: SDF ≤ repo ≤ MSF ‡

### Money market instruments ★★★★☆

| Instrument | Feature |
|---|---|
| Call / notice / term money | 1 day / 2–14 days / 15 days–1 yr interbank |
| T-bills | 91/182/364 days, discount, GoI |
| Cash management bills | <91 days, GoI |
| CP | Companies, 7 days–1 yr |
| CD | Banks, 7 days–1 yr |
| Repo / TREPS | Collateralised borrowing |
| MIBOR | Mumbai interbank overnight rate (benchmark for OIS) |

### Government securities market ★★★☆☆

- Dated G-secs, T-bills, SDLs, STRIPS
- RBI manages issuance (auctions: uniform price / multiple price)
- Primary dealers underwrite
- NDS-OM trading platform; SGL accounts
- RBI Retail Direct for individuals
- Zero credit risk; benchmark for other rates

### Bond features ★★★★☆

| Feature | Meaning |
|---|---|
| Face value | Par, repaid at maturity (₹100 quote basis) |
| Coupon | Annual interest % of face |
| Maturity | Redemption date |
| Zero-coupon | No coupon, issued at discount |
| Floating rate | Coupon resets to benchmark |
| Call / put option | Issuer can redeem early / investor can sell back |
| Credit rating | AAA highest; below BBB− = junk |

### Bond pricing ★★★★★

```
Price = Σ C/(1+y)^t + FV/(1+y)^n
Dirty (full) price = Clean price + Accrued interest
Price ↑ when yield ↓ (inverse)
Coupon > yield → premium bond · coupon < yield → discount
bond · equal → par
```

### Yield measures ★★★★★

| Measure | Formula / meaning |
|---|---|
| Coupon rate | Coupon ÷ face value |
| Current yield | Annual coupon ÷ market price |
| YTM | IRR of all cash flows held to maturity (assumes reinvestment at YTM) |
| YTC | Yield to call date |

### Duration, PVBP, convexity ★★★★★

```
Macaulay duration = weighted average time of cash flows
(years)
Modified duration  = MacD ÷ (1 + y/n)
%ΔP ≈ -MD × Δy           Ex: MD 6, y +0.25% → -1.5%
PVBP (DV01) = MD × Price × 0.0001
Convexity: price rises more than it falls for same Δy
(helps holder)
```

| Duration higher when | Duration lower when |
|---|---|
| Longer maturity | Shorter maturity |
| Lower coupon | Higher coupon |
| Lower yield | Higher yield |
| Zero-coupon: MacD = maturity | |

### Yield curve & term structure ★★★★☆

| Shape | Meaning |
|---|---|
| Normal (upward) | Long > short rates |
| Inverted | Short > long; recession signal |
| Flat | Similar rates |
| Humped | Mid-tenor highest |

- Expectations theory
- Liquidity preference theory
- Market segmentation theory
- Steepening / flattening

### Corporate bond market ★★☆☆☆

- NCDs, PSU bonds
- Rated by CRISIL, ICRA, CARE, India Ratings
- Credit spread = corporate yield − G-sec yield

# 19 Interest Rate Derivatives (OTC)

≈2 marks · 4 subtopics · chapter priority ★★☆☆☆

## Forward Rate Agreement (FRA) ★★★☆☆

| | |
|---|---|
| **What** | Lock a borrowing/lending rate for a future period |
| **Settlement** | Only interest difference, discounted, paid at start of period |
| **Notation** | 3×6 FRA = 3-month period starting in 3 months |

```
Payment = (Ref rate - FRA rate) × Notional × days/365 ÷
(1 + Ref × days/365)
```

## Interest Rate Swap (IRS) ★★★☆☆

- Plain vanilla: fixed ↔ floating on notional (notional not exchanged)
- OIS: floating leg = overnight MIBOR (most common in India)
- MIFOR: MIBOR-forward premium based ‡
- Borrower fearing rate rise → pay fixed, receive floating

## Caps, floors, collars, swaptions ★★☆☆☆

| Product | Use |
|---|---|
| Cap | Max rate for floating borrower (series of call options on rate) |
| Floor | Min rate for floating lender |
| Collar | Buy cap + sell floor |
| Swaption | Option to enter a swap |

## Uses & regulation ★★☆☆☆

- Hedge, speculate, change liability/asset profile
- OTC IRD regulated by RBI
- Exchange IRD by RBI + SEBI

# 20 Exchange Traded Interest Rate Futures

≈10 marks · 8 subtopics · chapter priority ★★★★★

## Products & history (India) ★★★☆☆

- 2003: first IRF on NSE (low volumes)
- 2009: 10-yr GoI bond futures (physical, CTD basket)
- 2014 onwards: cash-settled single-bond futures ‡
- 91-day T-bill futures
- Traded on NSE, BSE

## Contract features ★★★★☆

| Item | Bond futures ‡ | T-bill futures ‡ |
|---|---|---|
| Underlying | GoI security (e.g. 6/10/13-yr) | 91-day T-bill |
| Contract size | ₹2 lakh face value | ₹2 lakh face value |
| Quote | Clean price per ₹100 | 100 − annualised discount yield |
| Settlement | Cash (daily & final) | Cash |
| Expiry | Monthly contracts (3 serial) | Monthly |
| MTM | Daily | Daily |

## Pricing of bond futures ★★★★☆

```
Futures price ≈ (Spot clean + accrued) × (1 + r t) -
coupons received (with interest) - accrued at expiry
Then ÷ conversion factor for the CTD (physical contracts)
```

- Cost of carry: financing cost vs coupon income
- Positive carry if coupon yield > repo

## CTD, conversion factor, invoice price ★★★★★

| Term | Meaning |
|---|---|
| Deliverable basket | Several bonds acceptable for delivery (physical contracts) |
| Conversion factor (CF) | Price of a bond at the notional yield (e.g. 7%) per ₹1; coupon > notional → CF > 1 |
| Invoice price | Futures settlement price × CF + accrued interest |
| CTD | Bond costing short least to deliver: min (price − F × CF); highest implied repo |
| Delivery options | Quality option (which bond) + timing option — with the short |
| Gross basis | Cash price − F × CF |
| Implied repo | Return from buy bond + sell futures + deliver |

```
Ex: F 98.50, CF 1.032, AI 1.20 → invoice = 98.50 × 1.032
+ 1.20 = 102.85
```

## Hedging with IRF ★★★★★

| Risk | Who | Trade |
|---|---|---|
| Rates ↑ (bond prices ↓) | Bond holder, bank, debt MF | Sell IRF |
| Rates ↓ | Future investor, insurer | Buy IRF |

```
N = (MD_p × V_p) ÷ (MD_f × V_f)        = PVBP_portfolio ÷
PVBP_futures
Duration change: N = (MD_target - MD_current) × V_p ÷
(MD_f × V_f)
Ex: PVBP 4,00,000 ÷ 160 = 2,500 contracts (sell)
```

## Speculation & arbitrage ★★★★☆

- Expect rates ↓ → buy IRF
- Expect rates ↑ → sell IRF / T-bill futures
- Cash-and-carry: buy bond + sell futures when implied repo > repo
- Reverse: short bond + buy futures when implied repo < repo

## Margins & settlement ★★★☆☆

- Initial margin (SPAN-based) + extreme loss margin
- Daily MTM in cash
- Final settlement on weighted average price ‡
- Clearing corporation guarantee

## Participants ★★☆☆☆

- Banks
- Primary dealers
- Mutual funds
- Insurance & pension funds
- FPIs (limits)
- Corporates
- Retail

# 21 Exchange Traded Interest Rate Options

≈≈5 marks · 4 subtopics · chapter priority ★★★☆☆

## Features ★★★★☆

| | |
|---|---|
| **Underlying** | GoI bond (price-based) ‡ |
| **Style** | European |
| **Settlement** | Cash |
| **Buyer** | Pays premium |
| **Seller** | Pays margin |

**TRAP** On a bond-PRICE option: call gains when rates fall; put gains when rates rise. On a RATE option (cap) it's the opposite.

## Payoffs & uses ★★★★☆

| Position | Gains when | Use |
|---|---|---|
| Long call (bond price) | Rates fall | Lock purchase price for future bond buyer |
| Long put (bond price) | Rates rise | Protect bond portfolio (floor) |
| Short call / put | Rates stable | Earn premium |

## Pricing ★★☆☆☆

- Black (1976) model for options on futures/bonds
- Inputs: price, strike, time, rate, volatility of bond price
- Volatility of bond price ↑ with duration

## Caps & floors as options ★★★☆☆

- Cap = series of caplets (calls on rate)
- Floor = series of floorlets (puts on rate)
- Collar = long cap + short floor

# 22 Strategies using Interest Rate Derivatives

≈≈5 marks · 4 subtopics · chapter priority ★★★★☆

## Hedging & duration management ★★★★★

| Goal | Trade |
|---|---|
| Cut duration / protect portfolio | Sell IRF |
| Add duration | Buy IRF |
| Floor on bond value | Buy puts on bond price |
| Lock future investment yield | Buy IRF / buy calls on bond |
| Floating borrower fears rise | Pay fixed in IRS / buy cap / sell IRF |
| Investor fears fall | Receive fixed in IRS / buy floor / buy IRF |

## Yield curve strategies ★★★★★

| View | Trade |
|---|---|
| Steepening (long yields ↑ more) | Buy short-tenor futures, sell long-tenor futures |
| Flattening (short yields ↑ more) | Sell short-tenor futures, buy long-tenor futures |
| Parallel shift | Outright long/short futures |

- Weight legs by PVBP so a parallel move nets to zero

## Basis & arbitrage trades ★★★☆☆

- Cash-and-carry (long basis): buy bond, sell futures
- Reverse cash-and-carry (short basis)
- Calendar spreads between IRF expiries
- Implied repo vs actual repo decides

## Asset-liability management ★★★☆☆

- Banks: fixed assets vs floating deposits → swap or sell IRF
- Insurers: long liabilities → buy duration
- Corporates: convert floating loan to fixed via IRS